How-To Library

Informatica

Integrating Informatica® Big Data Management 10.2.2 HF1 SP1 with Qubole

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Abstract

Customers of Amazon Web Services (AWS) and Informatica can integrate Big Data Management® 10.2.2 HF1 SP1 with Qubole, the data activation platform. This article describes how to prepare the Qubole and AWS environments and configure Big Data Management to run on Qubole clusters.

The integration requires you to apply EBF-16050 to the domain and to clients.

Supported Versions

- Big Data Management 10.2.2 HF1 SP1

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Overview

Qubole is a data activation platform that interacts with Amazon Web Services (AWS) to automate cloud computing tasks. Qubole and Informatica Big Data Management work together to enable advanced data management pipelines. Big Data Management can run on Qubole clusters to extract, transform, and load data to any of several data repositories on AWS.

To use Big Data Management with a Qubole cluster, you must configure various elements in the AWS and Qubole environments. Then you use the Qubole interface to configure Big Data Management to run on Qubole.

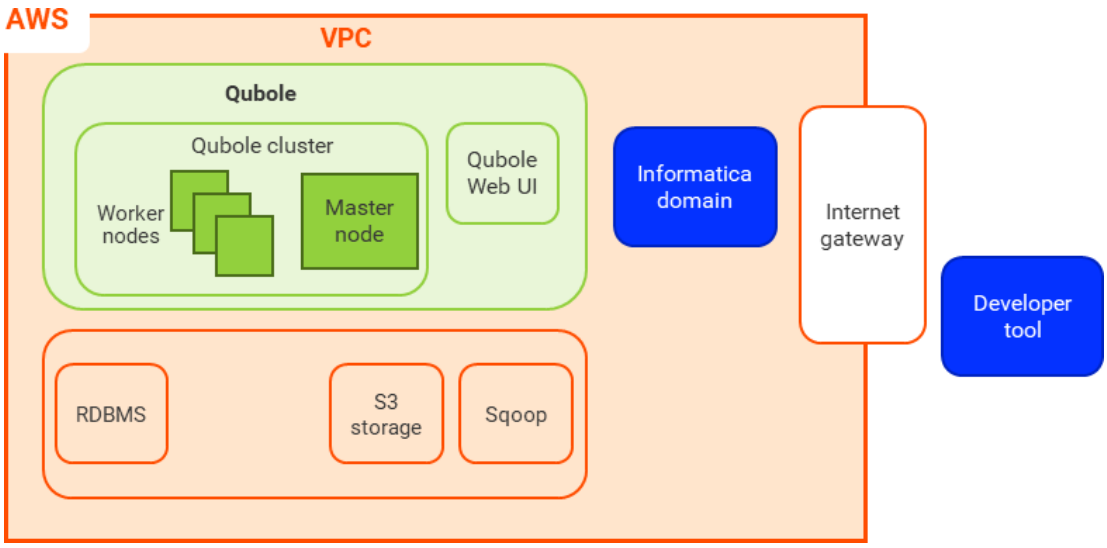
When configuration is complete, you can run mappings from the Developer tool in the Qubole environment on AWS.

Big Data Management Integration with Qubole

When you integrate Big Data Management with the Qubole implementation on AWS, you configure Qubole to communicate with Big Data Management.

Integrate Qubole with a deployment of Big Data Management in the same AWS virtual private cloud (VPC).

The following image shows the architecture of the Big Data Management integration with Qubole:

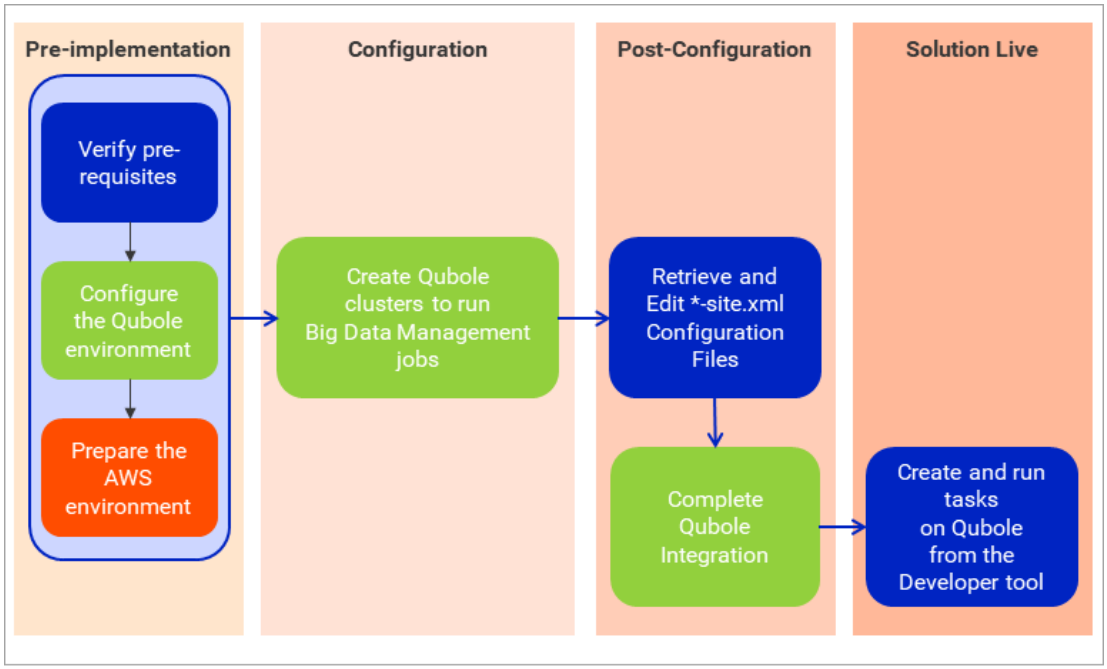


You can run Informatica clients on any machines inside or outside the AWS VPC.

The integration of Big Data Management with Qubole uses Informatica Power Exchange connectors to access and write data that resides in S3 or that is accessed via Sqoop.

Integration Overview

The following image shows how to integrate Big Data Management with Qubole on AWS:



Note: Kerberos configuration is not supported on the Qubole cluster with Big Data Management.

Informatica Domain

The Informatica domain is a server component that hosts application services, such as the Model Repository Service and the Data Integration Service. These services, together with domain clients, enable you to create and run mappings and other objects to extract, transform, and write data.

Application Services

Model Repository Service

The Model Repository Service is an application service that manages the Model repository. The Model repository stores metadata created by Informatica products in a relational database to enable collaboration among the products. Informatica Developer, the Data Integration Service, and the Administrator tool store metadata in the Model repository.

Data Integration Service

The Data Integration Service is an application service in the Informatica domain that performs data integration tasks for the Developer tool and for external clients.

Metadata Access Service

The Metadata Access Service is an application service that allows the Developer tool to access Hadoop connection information to import and preview metadata.

The Informatica domain can run other application services. For more information about Informatica services, see the *Informatica Application Service Guide*.

Domain Repositories

Informatica repositories, posted on SQL databases, store metadata about domain objects. Informatica repositories include the following:

Domain configuration repository

The domain configuration repository stores configuration metadata about the Informatica domain. It also stores user privileges and permissions.

Model repository

The Model repository stores metadata for projects and folders and their contents, including all repository objects such as mappings and workflows.

Monitoring Model repository

The monitoring Model repository stores statistics for Data Integration Service jobs. You configure the monitoring Model Repository Service in the domain properties.

In addition to these domain repositories, the Informatica deployment also requires a repository for Hive metadata. This repository is hosted on an SQL database. It stores Hive table metadata to enable Hadoop operations.

For more information about domain repositories, see the *Informatica Application Service Guide*.

Informatica Clients

You can use the following clients with Informatica Big Data Management:

Administrator tool

The Administrator tool enables you to create and administer services, connections, and other domain objects.

Developer tool

The Developer tool enables you to create and run mappings and other objects that enable you to access, transform, and write data to targets.

Command line interface

The command line interface offers hundreds of commands to assist in administering the Informatica domain, creating and running repository objects, administering security features, and maintaining domain repositories.

Qubole Clusters

Big Data Management supports a subset of cluster types, node types, and other features that are part of Qubole.

Qubole cluster types

Qubole allows you to create several types of clusters. Big Data Management integration supports the Hadoop Qubole cluster type.

Hadoop Qubole cluster node types

Qubole uses Amazon EC2 instances for cluster nodes. Each Hadoop Qubole cluster consists of a master node and at least one worker or task node, each of which runs a Hadoop distribution. Cluster nodes are of the following types:

Master node

Manages the cluster by running software components which coordinate the distribution of data and tasks among worker nodes for processing. The master node tracks the status of tasks and monitors the health of the cluster.

Worker node

Worker nodes are used for computing tasks. The worker nodes have software components that run tasks and store data in the Hadoop Distributed File System (HDFS) on the cluster.

Hadoop Qubole cluster data processing frameworks

The data processing framework layer is the engine used to process and analyze data. Many frameworks run on YARN or have their own resource management.

Informatica refers to these data processing frameworks as run-time engines. The engine you choose depends on processing needs, such as batch, interactive, in-memory, or streaming. Your choice of run-time engine affects the languages and interfaces on the application layer, which is the layer used to interact with the data you want to process.

Big Data Management can run jobs on the following run-time engines:

Spark

Apache Spark is a cluster framework and programming model for processing big data workloads. Like Hadoop MapReduce, Spark is an open-source, distributed processing system but uses directed acyclic graphs for execution plans and leverages in-memory caching for datasets. Spark supports multiple interactive query modules such as SparkSQL.

Note: By default, Qubole uses Spark version 2.1.1. The integration with Big Data Management requires Spark 2.2.1.

Blaze

Blaze is Informatica's unique data processing engine integrated with YARN to provide intelligent data pipelining, job partitioning, job recovery, and scalability, which is optimized to deliver high performance, scalable data processing leveraging Informatica’s cluster aware data integration technology.

AWS resources

Qubole uses AWS resources for the following elements:

EC2 Instances

Qubole uses Amazon EC2 virtual machines to host the Qubole server and web client as well as Qubole cluster nodes. When you configure Qubole integration with Big Data Management, you select from available EC2 node types. the node types available as Qubole cluster nodes are the same types as listed in AWS documentation.

Cluster resource management

The resource management layer is responsible for managing cluster resources and scheduling the jobs for processing data. By default, AWS uses YARN, which is a component introduced in Apache Hadoop 2.0 to centrally manage cluster resources for multiple data-processing frameworks.

Pre-Integration Tasks

Before you start to configure Big Data Management for use with Qubole, perform the following pre-integration tasks:

- Verify prerequisites.
- Perform initial Qubole configuration tasks.
- Prepare the AWS environment.

Verify Prerequisites

Verify the following prerequisites before you begin to configure Big Data Management for use with Qubole:

- Your AWS account has permissions to create resources, rules, and policies.

- Qubole is installed in your AWS environment.
- You have permissions to configure Qubole for the Big Data Management integration.

Perform Initial Qubole Configuration Tasks

Before you begin to integrate Big Data Management with Qubole on AWS, perform initial tasks to configure Qubole.

Establish the Qubole Environment on AWS

Go to the Qubole AWS Quick Start Guide and prepare the Qubole environment by performing the steps under the heading "Getting Started with Qubole on AWS."

Enable Spark 2.2.1 on Qubole

By default, Qubole supports Spark version 2.1.1, but the integration with Big Data Management requires Spark 2.2.1. To use Spark version 2.2.1, you must contact Qubole Support to enable it for your Qubole environment.

Create a support ticket with Qubole Support and request the following changes to your Qubole environment:

- Enable Spark 2.2.1 support for resources that you create.
- Disable private IP.
  Big Data Management uses VM host names to communicate with the Qubole cluster. When private IP is disabled for the cluster, the cluster identifies its nodes internally by host names, instead of IP addresses. This enables the Qubole cluster to understand resource requests from Informatica.
- Set the Hive Metastore version to 1.2.

Prepare the AWS Environment

Before you create Qubole clusters to run Big Data Management jobs, perform the steps in this section to prepare the AWS environment.

Deploy Big Data Management on AWS

You can integrate Qubole with a deployment of Big Data Management in the AWS cloud environment or in an on-premises environment.

If you want to implement a cloud deployment of Big Data Management, perform the following steps to deploy Big Data Management on AWS:

1. In the same subnet of the AWS VPC where you plan to create the Qubole cluster, create an Amazon EC2 instance to host the Informatica domain.

   When you create an EC2 instance, you choose from among the available EC2 instance types. For the domain VM, Informatica recommends a minimum of 64 GB RAM and a 16-core CPU. The m4.4xlarge instance type fulfills these requirements.
2. Configure the domain VM to be a member of the same security group as Qubole.
3. Upload the Big Data Management license and Informatica binaries to the domain EC2 instance, and run the big data installer.

   See the Informatica Big Data Installation and Configuration Guide.
4. Apply the Qubole enablement patch EBF-16050 to Informatica domain and client machines.

   Important: To enable support for Qubole, Big Data Management requires you to apply the patch on each of the EC2 instances that host Informatica domain and client nodes.

Download and Apply the EBF

Before you can download the EBF, you must request Informatica Global Customer Support to add you to a list of users entitled to download hot fix patches.

After your account is added, follow these steps to download the EBF patch:

1. Log in to https://network.informatica.com
2. Click **HotFix Downloads**.

   Note: You must log in to the site to see the HotFix Downloads link.
3. Use the search box to find an EBF. For example, enter EBF-16050.
4. Download the EBF archive file, which has a filename with the following format: `EBF-<number>.<os>.tar.gz`. For example: `16050.Linux64-X86.tar.gz`.

Create and Use an AWS Security Group

Create an AWS security group to secure communications between the domain, Qubole, and AWS, and use it in domain, cluster, and VPC configuration.

1. Create an AWS security group for the Big Data Management on Qubole implementation.

   Configure the security group with the following permissions:

   - To enable the cluster to use SSH to communicate with Informatica domain nodes, open all traffic for SSH.
   - For the domain host EC2 instance, enable all TCP communication for requests from domain clients. To determine the IP address of a client VM, log into the VM and use the command `whatsmyip`.

   See AWS documentation.
2. Provide the name of the security group in the "Persistent Security Groups" property in the Advanced properties panel when you run the Qubole cluster creation wizard.

Open Ports

Open ports on AWS to enable communication with the Informatica domain for Hadoop services.

Open the ports only for the Data Integration Service user.

The following table lists the ports to open:

Hadoop Component	Port
HDFS read/write	50010, 8010
HDFS namenode HTTP	50070
Hive metastore	10000
HiveServer	10003
JobTracker	9001
JobTracker web manager	50030
MapReduce Application Master	50100-50200
MapReduce JobHistory server	10020

Hadoop Component	Port
MapReduce JobHistory server webapp	19888
NameNode	9000
ResourceManager	8032
ResourceManager webapp	8088
ResourceTracker	8031
Scheduler address	8030
Shuffle HTTP	13562
TaskTracker web management	50060
ZooKeeper	2181

Generate RSA Keys

RSA security keys enable the domain and Qubole to communicate using SSH.

When you create a security key file on the domain and configure the Qubole cluster with the security key, Qubole cluster nodes accept traffic coming from the domain IP address.

1. Open a shell using Putty or another utility.
2. Use commands to generate an RSA key file.
   a. Type the following command to generate RSA keys:
      ```
      ssh-keygen -t rsa
      ```
   b. In the prompt, enter a file path. Use the following template:
      ```
      /Users/user_name/rsakey
      ```
   c. Enter and confirm a password.
   d. Make a note of the Key Fingerprint from the terminal output.
3. Navigate to the `rsakey` folder that you created in step 2b. Open the `rsakey.pub` file for editing.
4. Locate the public key beginning with *ssh* and copy the key.

   Use the public key to populate the Customer Public SSH Key property in the Advanced properties screen of the Qubole cluster when you create the cluster. See the Security Setting section in Create Qubole Clusters to Run Big Data Management Jobs.
5. Close the file and verify that the file permissions are set to be readable only by the account that you named in the key generation command.
   - For Linux systems, type `chmod 700`.
   - For Windows systems, follow the instructions in the following link: https://superuser.com/questions/1296024/windows-ssh-permissions-for-private-key-are-too-open.

Download the JDBC Drivers for Sqoop Connectivity

You can use the JDBC driver to configure connectivity to JDBC-compliant databases. You can use the JDBC connection to run mappings on the Spark engine. You can also enable Sqoop in the JDBC connection to run mappings on the Blaze and Spark engines.

You can use the DataDirect drivers shipped with Informatica to import metadata from JDBC-compliant databases. If a DataDirect driver for a specific database is not available, download the JDBC driver that the database vendor recommends. The DataDirect JDBC drivers that Informatica ships are not licensed for Sqoop mappings. To run JDBC mappings on the Spark engine or to run Sqoop mappings on the Blaze or Spark engine, use the JDBC driver that the database vendor recommends.

1. To connect to sources, targets, or lookups from JDBC-complaint databases, download any Type 4 JDBC driver that the database vendor recommends.
2. To import metadata in the Developer tool, place the JDBC driver .jar files in the following directory on the Developer tool machine:

   `<Informatica installation directory>\clients\externaljdbcjars`
3. To optimize the Sqoop mapping performance on the Spark engine while writing data to an HDFS complex file target of the Parquet format, download the following .jar files:
   - `parquet-hadoop-bundle-1.6.0.jar` from http://central.maven.org/maven2/com/twitter/parquet-avro/1.6.0/
   - `parquet-avro-1.6.0.jar` from http://central.maven.org/maven2/com/twitter/parquet-hadoop-bundle/1.6.0/
   - `parquet-column-1.5.0.jar` from http://central.maven.org/maven2/com/twitter/parquet-column/1.5.0/
4. To run mappings, copy all of the .jar files to the following directory on the machine where the Data Integration Service runs:

   `<Informatica installation directory>\externaljdbcjars`

   Changes take effect after you recycle the Data Integration Service. At run time, the Data Integration Service copies the .jar files to the Hadoop distribution cache so that the .jar files are accessible to all nodes in the cluster.

Setting S3 Access Policies

The AWS administrator must set S3 access policies to grant users the required access to S3 resources.

S3 access policies allow control of user access to S3 resources and the actions that users can perform. The AWS administrator uses policies to control access and actions for specific users and resources, depending on the use case that mappings and workflows require.

AWS uses a JSON statement for S3 access policies. To set the S3 access policy, determine the principal, actions, and resources to define, then create or edit an existing S3 access policy JSON statement.

For more information about Amazon S3 access policies, see AWS documentation.

Step 1. Identify the S3 Access Policy Elements

Identify the principal, actions, and resources to insert in the access policy.

The following table describes the tags to set in the access policy:

Tag	Description
Principal	The user, service, or account that receives permissions that are defined in a policy. Assign the owner of the S3 bucket resources as the principal. **Note:** The S3 bucket owner and the owner of resources within the bucket can be different.
Action	The activity that the principal has permission to perform. In the sample, the Action tag lists two put actions and one get action. You must specify both get and put actions to grant read and write access to the S3 resource.
Resource	The S3 bucket, or folder within a bucket. Include only resources in the same bucket.

Sample S3 Policy JSON Statement

The following JSON statement contains the basic elements of an S3 bucket access policy:

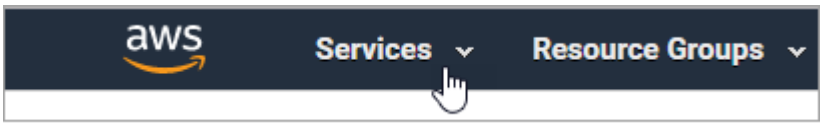
```
{
"Version": "<date>",
"Id": "Allow", "Statement": [
{ "Sid": "<Statement ID>", "Effect": "Allow",
"Principal": {
                "AWS": "arn:aws:iam::<account_2_ID>:<user>"
}
"Action":[
"s3:PutObject","s3:PutObjectAcl",
"s3:GetObject"
]
"Resource": [
"Resource": "arn:aws:s3:::<bucket_1_name>/foldername/*"
]
}
```

Step 2. Optionally Copy an Existing S3 Access Policy as a Template

When the AWS administrator selects a role for cluster users, the AWS console generates a default access policy. After the AWS console generates the default policy, you can copy it and customize it to grant access to specific resources to specific users.

Complete the following steps to copy an existing S3 access policy:

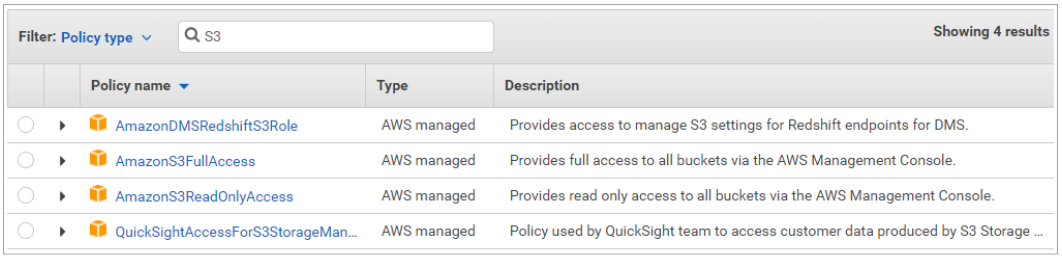
1. In the AWS console, click the **Services** menu.
   The image below shows the **Services** menu in the menu bar:



2. Type "IAM" in the search bar and press Enter.
   The **Welcome to Identity and Access Management** screen opens.
3. In the menu on the left, select **Policies**.
   The console displays a list of existing policies.

4. Type "S3" in the search bar and press Enter.
   The console displays a list of existing S3 access policies.

   The image below shows an example of a list of S3 access policies:



5. Click the name of the policy that you want to copy.
   The policy opens in a read-only window.
6. Highlight and copy the policy statement.

After you copy the JSON statement, you can edit it in a text editor or in the bucket policy editor.

Step 3. Create or Edit an S3 Access Policy

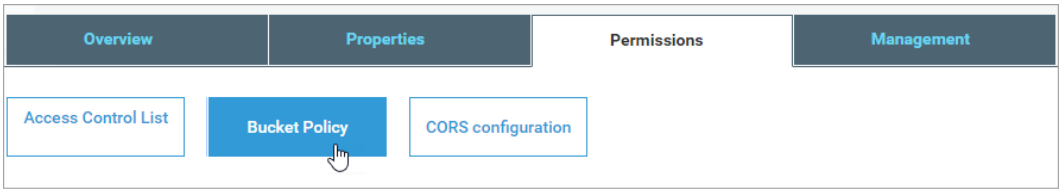
Create an S3 access policy or edit an existing policy. The AWS administrator can enter a JSON statement, based on a template. The administrator can copy and customize the S3 policy from another bucket.

1. In the AWS console, click the **Services** menu.
2. In the **Storage** section, choose **S3**.

   The AWS console displays a list of existing buckets.
3. Use the **search box** to find the bucket you want to set a policy for, and select the bucket from the results.
4. Click the **Permissions** tab, then click **Bucket Policy**.

   The **Bucket Policy Editor** opens.

   The image below shows the **Bucket Policy** button:



5. Type the bucket access policy, or edit the existing policy, and click **Save**.

   AWS applies the access policy to the bucket.

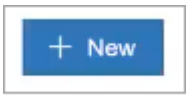
Create Qubole Clusters

Create and configure Qubole clusters to run Big Data Management jobs.

Launch the Cluster Creation Process

To create a Qubole cluster to run Big Data Management jobs, perform the following steps in the Qubole web console:

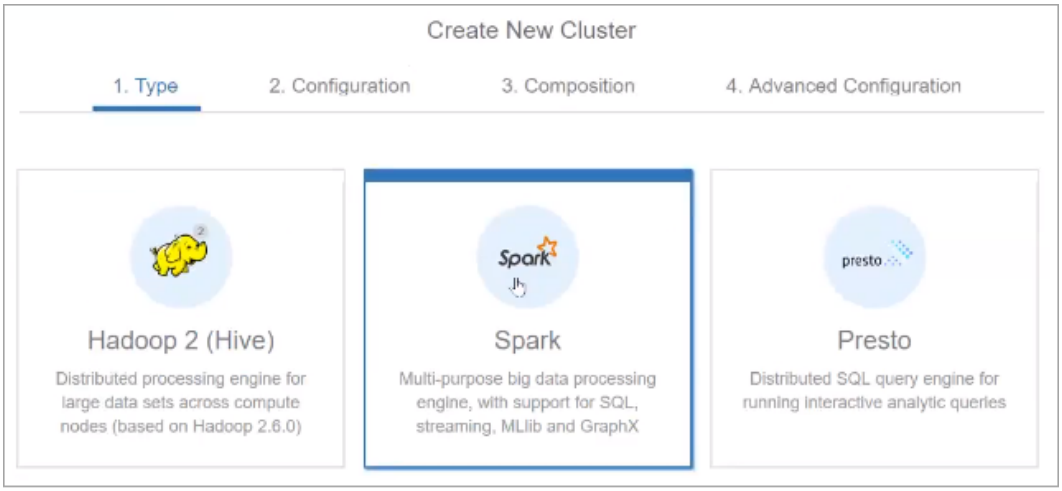
1. On the **Clusters** page, click the **New** button. The following image shows the **New** button:



   The first panel of the cluster configuration interface opens.

2. Choose the Spark type cluster.

   The following image shows the Spark cluster choice:



   The next configuration panel opens.

Configure the Cluster

In the next panel of the cluster creation wizard, you configure cluster settings.

1. Configure cluster settings.

   The following table describes the settings to configure:

Property	Description
Cluster Labels	Specify a string to use as a prefix to cluster node host names.
Spark Version	Select `2.2 latest -- Spark 2.2.1`. **Note:** You must perform separate configuration steps to enable Spark version 2.2.1. See the following topics in this article: - "Perform Initial Qubole Configuration Tasks" on page 7 - "Deploy Big Data Management on AWS " on page 7

Property	Description
Notebook Interpreter Node	Qubole protocol for using Qubole notebooks. Accept the default setting. **Note:** Big Data Management version 10.2x does not support integration with Qubole notebooks.
Master Node Type	Select an EC2 instance type from the drop down list. For the master node, Informatica requires a minimum 16 cores and 64 GB memory.
Worker Node Type	Select an EC2 instance type from the drop down list. For the worker node, Informatica requires a minimum 16 cores and 64 GB memory.
Use Multiple Worker Node Types	Select this option if you want to specify multiple EC2 instance types for worker nodes.
Minimum Worker Nodes	Specify the minimum number of worker nodes to provision.
Maximum Worker Nodes	Specify the maximum number of worker nodes to provision.
Region	Specify the AWS region in which to create the cluster.
Availability Zone	Availability zone where the cluster is deployed. Accept the default value. Default: Any.
EBS Volume Count	Count of EBS (elastic block storage) volumes to be mounted to an instance as reserved disks. Default: 0
EBS Volume Type	Type of EBS volume to be mounted to an instance as reserved disk. Choose from the following options: - ssd (gp2 SSD). General purpose solid state drive. - standard (standard HDD). Standard-sized hard disk drive. - st1 (HDD). Throughput-optimized hard disk drive. - sc1 (HDD). Cold hard disk drive. For more information about EBS volume types, see Amazon documentation.
EBS Volume Size	Size (in GB) of each EBS volume to be mounted to an instance as reserved disk.
Enable EBS Upscaling	Enable dynamic block storage up-scaling. Default is disabled.
Node Bootstrap File	Name of the file that contains bootstrap instructions to run after the Qubole cluster is provisioned. **Note:** The file must be in the default path displayed above the text entry pane. For more information about node bootstrap files, see Qubole documentation.
Disable Automatic Cluster Termination	Check this option to disable Qubole automatic cluster termination. Informatica requires disabling of automatic termination.
Idle Cluster Timeout	Applies only to automatic cluster termination. Disabled if automatic cluster termination is disabled.

2. Click **Next** to continue to the next panel.

In the next panel of the cluster creation wizard, you can configure cluster composition. It is not necessary to change any of the default settings in this panel.

3. Click **Next** to continue to the next panel, where you configure advanced properties.

Configure Advanced Properties

In the next panel of the cluster creation wizard, you configure advanced properties.

EC2 Settings

The following table describes EC2 settings to configure:

Property	Description
Same as Default Compute	Check this option if you want the cluster to use the default compute settings for the user account. Default is selected.
VPC	Click the dropdown and select the AWS VPC where the Informatica domain is deployed. Default: None.
Subnet	Click the dropdown and select the AWS VPC subnet where the Informatica domain is deployed. Default: None.
Elastic IP for Master Node	Supply the elastic IP address of the cluster master node.
Custom EC2 Tags	Type a string to identify cluster nodes that are part of the Informatica integration. To enable reporting, add the following tag: `Informatica_BDM_10.2.2`

Hadoop Cluster Settings

The following table describes Hadoop cluster settings to configure:

Property	Description
Override Hadoop Configuration Variables	Use the text field to enter custom variables to enable Big Data Management integration. When you enter values in this field, Qubole updates the *-site.xml configuration files on each cluster node. For the integration with Big Data Management, enter the following key-value pair to set the YARN node manager timeout: `yarn.resourcemanager.app.timeout.minutes=-1`

Do not enter values for any of the other settings in this section.

Spark Settings

It is not necessary to change any of the settings in this section.

Monitoring

It is not necessary to change any of the settings in this section.

Security Settings

The following table describes security settings to configure:

Property	Description
Qubole Public Key	Qubole automatically populates this field with a public key.
Customer SSH Public Key	Public key for the Informatica domain VM. Qubole requires this key to establish communication between the Qubole cluster and the domain. Enter the SSH public key that you created previously. The value is stored in the file `id_rsa.pub` in the directory on the Big Data Management domain node that you created when you created the RSA keys. See "Generate RSA Keys " on page 9.
Persistent Security Groups	Enter the name of the IAM security group that you created for the Qubole-Big Data Management integration.

Do not enter values for any of the other settings in this section.

Click **OK** to complete cluster creation.

Enable a Qubole Spark Cluster to Use Hive Queries

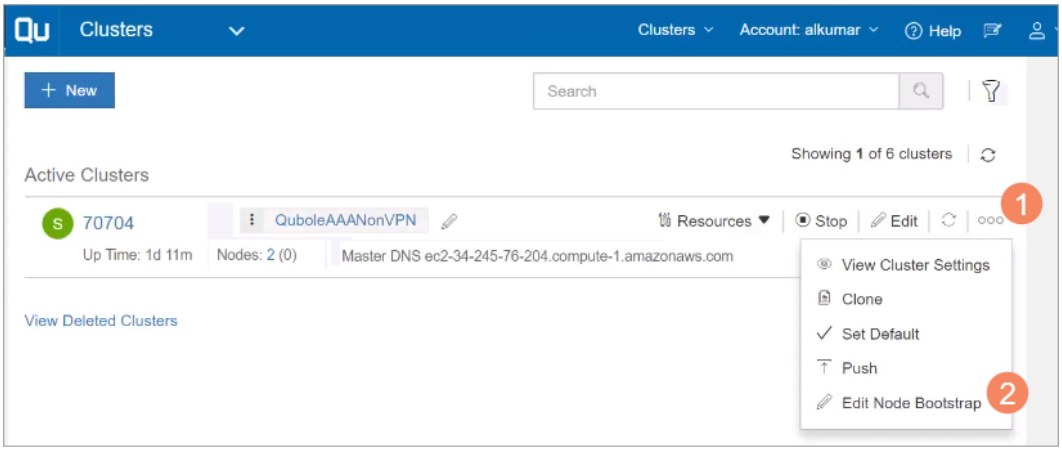
Big Data Management runs on the Qubole Spark type cluster. The Qubole spark cluster does not enable Hive queries by default.

To enable Big Data Management to process Hive queries, use a bootstrap script file to perform additional configuration on the Qubole cluster after provisioning. Get the bootstrap script from Qubole Support. The bootstrap script file is named `hiveserver2-admin`.

After you procure the bootstrap script file, perform the following additional cluster configuration steps:

1. Upload the bootstrap script to the following path on the Qubole server: `/usr/lib/hive1.2/bin/`.
2. Select the cluster you created to run Big Data Management.
3. Click the edit icon for the cluster.

   The following image shows the edit icon, which looks like three small dots, and is marked with the numeral 1:



   The Edit menu opens.

4. Choose **Edit Node Bootstrap**.

A command shell for the Qubole server opens.

5. Enter the following command to launch the bootstrap script:

```
/usr/lib/hive1.2/bin/hiveserver2-admin start
```

The bootstrap script configures the Qubole server to accept Hive queries from Big Data Management.

You can use the edit node bootstrap window for other commands to configure the Qubole cluster after you have provisioned and started it.

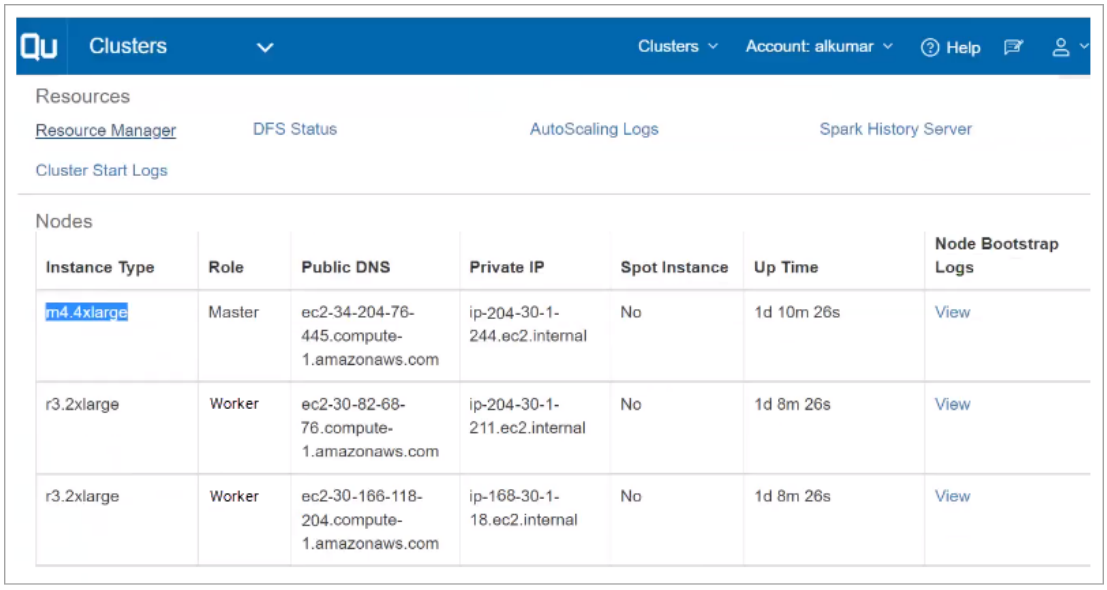
Monitoring Qubole Cluster Provision

You can use cloud platform dashboards, logs, or other artifacts to see whether cluster creation succeeded and how to locate and identify the Informatica domain on the cloud platform.

Qubole Cluster Resources

Use the clusters dashboard in the Qubole portal to see a list of nodes in the cluster.

The following image shows a cluster with two worker nodes:



Logs

Click the Logs tab in the Administrator tool to view the installation logs that the automated deployment created.

Installation log

Records the installation of the Informatica domain, Informatica services, and repositories. In case any of the installation tasks failed, you can check the log for the command that failed.

Filename: `Informatica_<version>_Services_<timestamp>.log`

Location: `/home/Informatica`

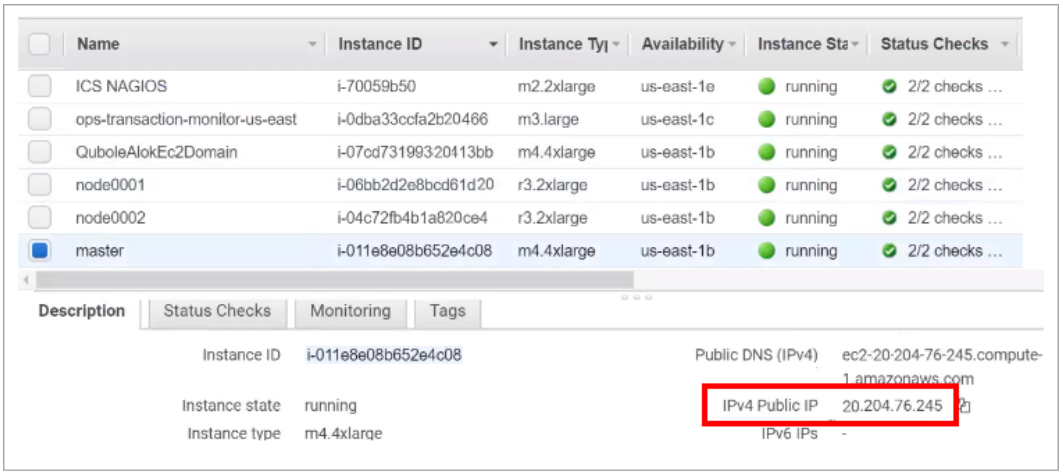
Edit *-site.xml Cluster Configuration Values

Big Data Management uses the property-value pairs from a set of cluster configuration files on the Qubole cluster to create a cluster configuration object. The Data Integration Service uses the cluster configuration to run mappings on the cluster.

To create a cluster configuration, you must retrieve the set of cluster configuration files from the Qubole cluster and save them to the domain node.

1. In the AWS console, browse to the list of running EC2 instances. Select the instance that hosts the Qubole master node.
2. In the Description tab of the EC2 instance, copy the value of the IPv4 Public IP property.

   The following image shows the location of the IPv4 Public IP property:



3. Open a shell on the Informatica domain host using Putty or a similar utility.
4. Type the following command to open an SSH channel to the Qubole cluster master node: `ssh ec2-user@<IPv4 address>`
5. Browse to the Hadoop configuration directory on the cluster master node: `cd /etc/hadoop`

   When you list the files in the Hadoop configuration directory, note the presence of several files named like *-site.xml. These are the Hadoop cluster configuration files.
6. **Note:** In this step and the next step, you copy files from the cluster master node to any directory on the Informatica domain node.

   Copy the following files from the cluster master node to a directory on the domain node:

   - core-site.xml
   - hdfs-site.xml
   - mapred-site.xml
   - yarn-site.xml
7. Browse to the `usr/lib/hive1.2/conf` directory and copy the hive-site.xml file to the directory on the domain node where you copied the other *-site.xml files.
8. Get the fully qualified host name of the Informatica domain node. To get this value, type the command `hostname`.

   The resulting value will be like `ip-10-20-30-40.ec2.internal`.

9. On the Informatica domain node directory where you copied the *-site.xml files, open the `core-site.xml` file for editing. Make the following change:
   a. Locate the property `fs.defaultFS`.
   b. Change the value from: `hdfs://${master.hostname}:9000` to contain the fully qualified host name from the previous step.

      For example, the resulting value will be like `hdfs://ip-10-20-30-40.ec2.internal:9000`.
   c. Save and close the `core-site.xml` file.
10. Open the `hive-site.xml` file for editing. Make the following change:
   a. Copy the following property entry and paste it to the file:

```
<property>
   <name>hive.metastore.uris</name>
   <value>thrift://<fully qualified master host name>:10000</value>
   <description>JDBC connect string for a JDBC metastore</description>
</property>
```

   b. Replace `<fully qualified master host name>` with the fully qualified host name from the previous step.

      For example, the resulting value will be like `thrift://ip-10-20-30-40.ec2.internal:10000`.
   c. Save and close the `hive-site.xml` file.
11. Create a .zip archive containing the five *-site.xml files.

    Use this .zip archive file to create the cluster configuration on the domain.

Complete Quoble Integration

Perform the tasks in this section to complete integration of Qubole with Big Data Management.

Edit the hosts File for the Blaze Engine

To run the Blaze engine on every node in the cluster, verify that the /etc/hosts file on every node has entries for all other nodes.

Each node in the cluster requires an entry for the IP address and the fully qualified domain name (FQDN) of all other nodes. For example,

```
127.0.0.1 localhost node1.node.com
208.164.186.1 node1.node.com node1
208.164.186.2 node2.node.com node2
208.164.186.3 node3.node.com node3
```

Changes take effect after you restart the network.

Create a Cluster Configuration

After the Hadoop administrator prepares the cluster for import, the Informatica administrator must create a cluster configuration.

A cluster configuration is an object in the domain that contains configuration information about the Hadoop cluster. The cluster configuration enables the Data Integration Service to push mapping logic to the Hadoop environment.

Usually, the Informatica administrator imports configuration properties directly from the Hadoop cluster to create a cluster configuration. In the Qubole integration process, you have already copied the *-site.xml configuration files to a directory on the domain node (see "Edit *-site.xml Cluster Configuration Values" on page 18). The import process imports values from these *-site.xml files into configuration sets that correspond to the individual *-site.xml files.

The cluster configuration wizard can create Hadoop, HDFS, and Hive connections to enable the Informatica domain to access the Hadoop environment. If you choose to create the connections, the wizard also associates the cluster configuration with the connections.

For more information about the cluster configuration, see the *Data Engineering Administrator Guide*.

1. From the **Connections** tab, click the **ClusterConfigurations** node in the Domain Navigator.
2. From the Actions menu, select **New** > **Cluster Configuration**.

   The **Cluster Configuration** wizard opens.
3. Configure the following properties:

Property	Description
Cluster configuration name	Name of the cluster configuration.
Description	Optional description of the cluster configuration.
Distribution type	The cluster Hadoop distribution type. Select Qubole.
Distribution version	Version of the Hadoop distribution. Each distribution type has a default version. This is the latest version of the Hadoop distribution that Big Data Management supports. When the cluster version differs from the default version, the cluster configuration wizard populates the cluster configuration Hadoop distribution property with the most recent supported version relative to the cluster version. For example, suppose Informatica supports versions 5.10 and 5.13, and the cluster version is 5.12. In this case, the wizard populates the version with 5.10. You can edit the property to choose any supported version. Restart the Data Integration Service for the changes to take effect.
Method to import the cluster configuration	Choose **Import from file** to import properties from an archive file.
Create connections	Choose to create Hadoop, HDFS, Hive, and HBase connections. If you choose to create connections, the Cluster Configuration wizard associates the cluster configuration with each connection that it creates. If you do not choose to create connections, you must manually create them and associate the cluster configuration with them. **Important:** When the wizard creates the Hive connection, it populates the Metadata Connection String and the Data Access Connection String properties with the value from the hive.metastore.uris property. If the Hive metastore and HiveServer2 are running on different nodes, you must update the Metadata Connection String to point to the HiveServer2 host.

4. Click **Browse** to select a file. Select the file and click **Open**.
5. Click **Next** and verify the cluster configuration information on the summary page.

Configure the Informatica Domain

Perform the tasks in this section to configure settings in the domain.

Before you begin these tasks, you must have created the cluster configuration for the Qubole cluster.

Edit core_site_xml Settings in the Cluster Configuration

Perform this task to enable the Data Integration Service to run mappings that access resources on S3 storage.

Use the Administrator tool to add the following property-value pairs to the core_site_xml configuration set in the cluster configuration:

- `fs.s3.isClusterOnEc2Role=FALSE`
- `fs.s3n.isClusterOnEc2Role=FALSE`
- `fs.s3n.awsSecretAccessKey=<secret access key value>`
- `fs.s3n.awsAccessKeyId=<access key id value>`
- `fs.s3a.buffer.dir=/tmp`

Edit Data Integration Service Properties

To access Sqoop sources and targets with mappings that use the Spark engine, edit the Data Integration Service Execution Options to add the following line:

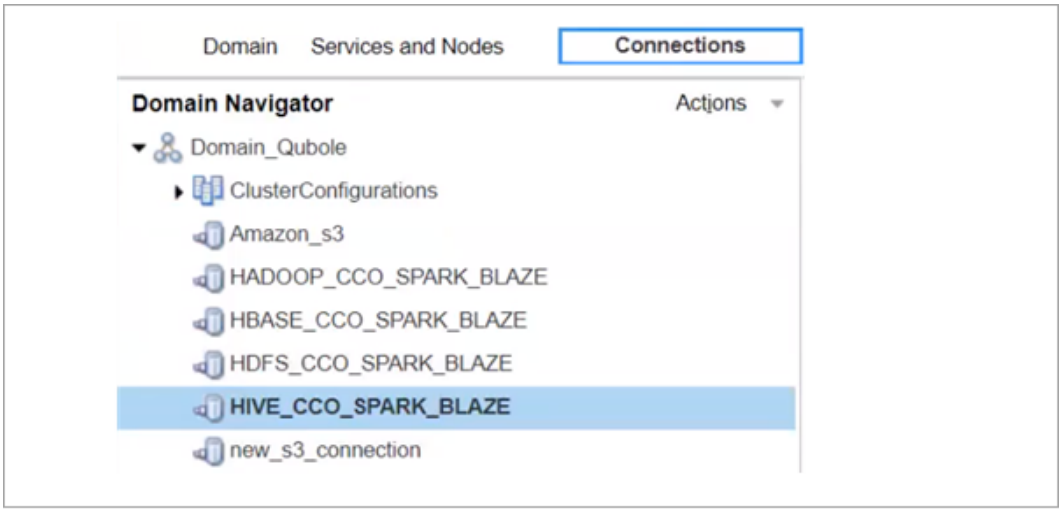
```
SparkSqoopDisSideInvocation=false
```

Configure the Hadoop Connection

Hadoop connections contain default values for properties such as cluster environment variables, cluster path variables, and advanced properties. Based on the cluster environment and the functionality that you use, you can add to the default values or change the default values of these properties.

After you create Hadoop connections during cluster configuration, use the Administrator tool to edit the Hive connection to change the connection information for the Hive metadata repository:

1. In the Administrator tool, expand the list of connections.
   The image below shows an example of the list of connections in the domain navigator:



   The cluster configuration wizard creates connections and names them using the name of the cluster configuration as a suffix identifier. Each connection name begins with a connection type: HADOOP, HBASE, HDFS, and HIVE. In this image, the Hive connection is named HIVE_CCO_SPARK_BLAZE.
2. Open the HIVE connection.
3. In the Common Properties section of the connection properties, click the **Edit** icon to edit the properties.

4. Locate the Metadata Connection String property and change the port number to 10003.
5. Locate the Data Access Connection String property and change the port number to 10003.
6. Locate the Cluster Executable Path property and add the YARN executable path to the `$DEFAULT_CLUSTER_EXEC_PATH:`.

   Note: Run the `which yarn` command on the Qubole cluster to retrieve the YARN executable path. Add the YARN executable path till the `/bin` directory.

   For example, add the following path after the `$DEFAULT_CLUSTER_EXEC_PATH:` in the Cluster Executable Path property:

   ```
   /usr/lib/hadoop2/bin
   ```
7. Click **OK** to save the changes.

For a full list of Hadoop connection properties to configure, see "Hadoop Connection Properties " on page 23.

Configure the Developer Tool

Edit `developerCore.ini` to successfully import local complex files available on the Developer tool machine. Edit `developerCore.ini` on each Developer tool machine.

When you import a complex file, such as Avro or Parquet, the imported object includes metadata associated with the distribution in the Hadoop environment. If the file resides on the Developer tool machine, the import process picks up the distribution information from the `developerCore.ini` file. You must edit the `developerCore.ini` file to point to the distribution directory on the Developer tool machine.

You can find `developerCore.ini` in the following directory:

```
<Informatica installation directory>\clients\DeveloperClient
```

Add the following property:

```
-DINFA_HADOOP_DIST_DIR=hadoop\<distribution>_<version>
```

The change takes effect when you restart the Developer tool.

Configure developerCore.ini

Edit `developerCore.ini` to successfully import local complex files available on the Developer tool machine.

When you import a complex file, such as Avro or Parquet, the imported object includes metadata associated with the distribution in the Hadoop environment. If the file resides on the Developer tool machine, the import process picks up the distribution information from the `developerCore.ini` file. You must edit the `developerCore.ini` file to point to the distribution directory on the Developer tool machine.

You can find `developerCore.ini` in the following directory:

```
<Informatica installation directory>\clients\DeveloperClient
```

Add the following property:

```
-DINFA_HADOOP_DIST_DIR=hadoop\<distribution>_<version>
```

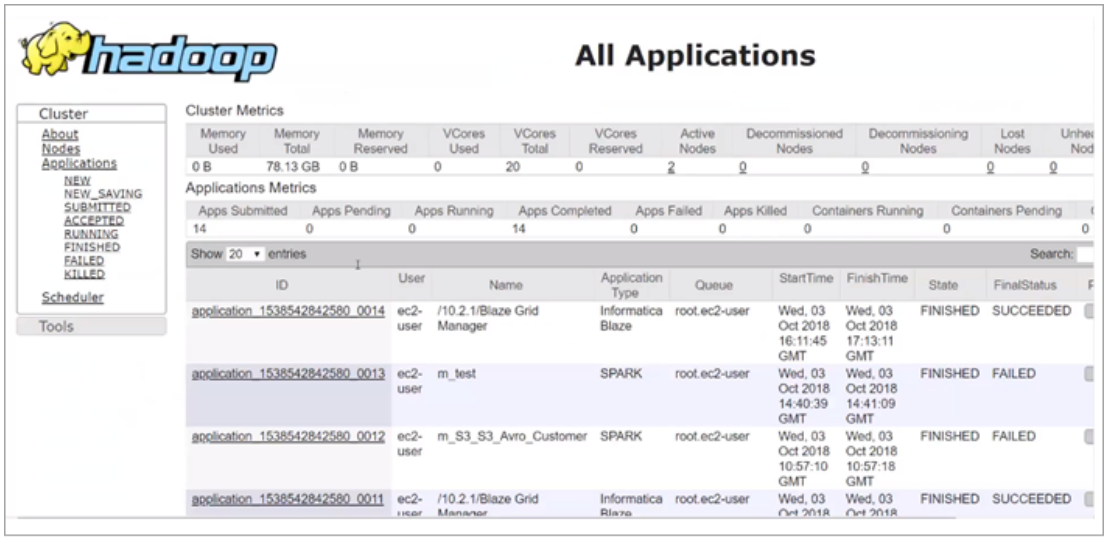
The change takes effect when you restart the Developer tool.

Monitoring Big Data Management Jobs

To monitor the status of Big Data Management jobs on the Qubole cluster, click **Resource Manager** on the Qubole Cluster Resources page.

When you click on the Resource Manager link, the Qubole portal displays the Hadoop Resource Manager. The Hadoop Resource Manager shows the status of all jobs that were sent to the Qubole cluster.

The following image shows the Hadoop Resource Manager:



Hadoop Connection Properties

Use the Hadoop connection to configure mappings to run on a Hadoop cluster. A Hadoop connection is a cluster type connection. You can create and manage a Hadoop connection in the Administrator tool or the Developer tool. You can use infacmd to create a Hadoop connection. Hadoop connection properties are case sensitive unless otherwise noted.

The following topics are included in this article for reference. Some of these properties are not required for the connections that the Qubole cluster configuration creates.

Hadoop Connection: General Properties

Configure properties in the Hadoop connection to enable communication between the Data Integration Service and the Hadoop cluster.

The following table describes the general connection properties for the Hadoop connection:

Property	Description
Name	The name of the connection. The name is not case sensitive and must be unique within the domain. You can change this property after you create the connection. The name cannot exceed 128 characters, contain spaces, or contain the following special characters: ~ ` ! $ % ^ & * ( ) - + = { [}] | \ : ; " ' < , > . ? /
ID	String that the Data Integration Service uses to identify the connection. The ID is not case sensitive. It must be 255 characters or less and must be unique in the domain. You cannot change this property after you create the connection. Default value is the connection name.
Description	The description of the connection. Enter a string that you can use to identify the connection. The description cannot exceed 4,000 characters.
Cluster Configuration	The name of the cluster configuration associated with the Hadoop environment. Required if you do not configure the Cloud Provisioning Configuration.
Cloud Provisioning Configuration	Name of the cloud provisioning configuration. **Note:** The feature that this property enables is not supported for the integration with Qubole.
Cluster Environment Variables*	Environment variables that the Hadoop cluster uses. For example, the variable ORACLE_HOME represents the directory where the Oracle database client software is installed. You can configure run-time properties for the Hadoop environment in the Data Integration Service, the Hadoop connection, and in the mapping. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Hadoop connection or in the mapping. The Data Integration Service processes property overrides based on the following priorities: 1. Mapping custom properties set using infacmd ms runMapping with the -cp option 2. Mapping run-time properties for the Hadoop environment 3. Hadoop connection advanced properties for run-time engines 4. Hadoop connection advanced general properties, environment variables, and classpaths 5. Data Integration Service custom properties
Cluster Library Path*	The path for shared libraries on the cluster. The $DEFAULT_CLUSTER_LIBRARY_PATH variable contains a list of default directories.

Property	Description
Cluster Classpath*	The classpath to access the Hadoop jar files and the required libraries. The $DEFAULT_CLUSTER_CLASSPATH variable contains a list of paths to the default jar files and libraries. You can configure run-time properties for the Hadoop environment in the Data Integration Service, the Hadoop connection, and in the mapping. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Hadoop connection or in the mapping. The Data Integration Service processes property overrides based on the following priorities: 1. Mapping custom properties set using infacmd ms runMapping with the -cp option 2. Mapping run-time properties for the Hadoop environment 3. Hadoop connection advanced properties for run-time engines 4. Hadoop connection advanced general properties, environment variables, and classpaths 5. Data Integration Service custom properties
Cluster Executable Path*	The path for executable files on the cluster. The $DEFAULT_CLUSTER_EXEC_PATH variable contains a list of paths to the default executable files.
* Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.	

Common Properties

The following table describes the common connection properties that you configure for the Hadoop connection:

Property	Description
Impersonation User Name	Required if the Hadoop cluster uses Kerberos authentication. Hadoop impersonation user. The user name that the Data Integration Service impersonates to run mappings in the Hadoop environment. The Data Integration Service runs mappings based on the user that is configured. Refer the following order to determine which user the Data Integration Services uses to run mappings: 1. Operating system profile user. The mapping runs with the operating system profile user if the profile user is configured. If there is no operating system profile user, the mapping runs with the Hadoop impersonation user. 2. Hadoop impersonation user. The mapping runs with the Hadoop impersonation user if the operating system profile user is not configured. If the Hadoop impersonation user is not configured, the Data Integration Service runs mappings with the Data Integration Service user. 3. Informatica services user. The mapping runs with the operating user that starts the Informatica daemon if the operating system profile user and the Hadoop impersonation user are not configured.
Temporary Table Compression Codec	Hadoop compression library for a compression codec class name. **Note:** The Spark engine does not support compression settings for temporary tables. When you run mappings on the Spark engine, the Spark engine stores temporary tables in an uncompressed file format.
Codec Class Name	Codec class name that enables data compression and improves performance on temporary staging tables.

Property	Description
Hive Staging Database Name	Namespace for Hive staging tables. Use the name `default` for tables that do not have a specified database name. If you do not configure a namespace, the Data Integration Service uses the Hive database name in the Hive target connection to create staging tables. When you run a mapping in the native environment to write data to Hive, you must configure the Hive staging database name in the Hive connection. The Data Integration Service ignores the value you configure in the Hadoop connection.
Advanced Properties	List of advanced properties that are unique to the Hadoop environment. The properties are common to the Blaze and Spark engines. The advanced properties include a list of default properties. You can configure run-time properties for the Hadoop environment in the Data Integration Service, the Hadoop connection, and in the mapping. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Hadoop connection or in the mapping. The Data Integration Service processes property overrides based on the following priorities: 1. Mapping custom properties set using infacmd ms runMapping with the `-cp` option 2. Mapping run-time properties for the Hadoop environment 3. Hadoop connection advanced properties for run-time engines 4. Hadoop connection advanced general properties, environment variables, and classpaths 5. Data Integration Service custom properties **Note:** Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.

Reject Directory Properties

The following table describes the connection properties that you configure to the Hadoop Reject Directory.

Property	Description
Write Reject Files to Hadoop	If you use the Blaze engine to run mappings, select the check box to specify a location to move reject files. If checked, the Data Integration Service moves the reject files to the HDFS location listed in the property, Reject File Directory. By default, the Data Integration Service stores the reject files based on the RejectDir system parameter.
Reject File Directory	The directory for Hadoop mapping files on HDFS when you run mappings.

Blaze Configuration

The following table describes the connection properties that you configure for the Blaze engine:

Property	Description
Blaze Staging Directory	The HDFS file path of the directory that the Blaze engine uses to store temporary files. Verify that the directory exists. The YARN user, Blaze engine user, and mapping impersonation user must have write permission on this directory. Default is `/blaze/workdir`. If you clear this property, the staging files are written to the Hadoop staging directory `/tmp/blaze_<user name>`.
Blaze User Name	The owner of the Blaze service and Blaze service logs. When the Hadoop cluster uses Kerberos authentication, the default user is the Data Integration Service SPN user. When the Hadoop cluster does not use Kerberos authentication and the Blaze user is not configured, the default user is the Data Integration Service user.
Minimum Port	The minimum value for the port number range for the Blaze engine. Default is 12300.
Maximum Port	The maximum value for the port number range for the Blaze engine. Default is 12600.
YARN Queue Name	The YARN scheduler queue name used by the Blaze engine that specifies available resources on a cluster. **Note:** If YARN preemption is enabled on the cluster, verify with the Hadoop administrator that preemption is disabled on the queue associated with the Blaze engine.
Blaze Job Monitor Address	The host name and port number for the Blaze Job Monitor. Use the following format: `<hostname>:<port>` Where - <hostname> is the host name or IP address of the Blaze Job Monitor server. - <port> is the port on which the Blaze Job Monitor listens for remote procedure calls (RPC). For example, enter: `myhostname:9080`
Blaze YARN Node Label	Node label that determines the node on the Hadoop cluster where the Blaze engine runs. If you do not specify a node label, the Blaze engine runs on the nodes in the default partition. If the Hadoop cluster supports logical operators for node labels, you can specify a list of node labels. To list the node labels, use the operators `&&` (AND), `||` (OR), and `!` (NOT). **Note:** You cannot use node labels on a Cloudera CDH cluster.
Advanced Properties	List of advanced properties that are unique to the Blaze engine. The advanced properties include a list of default properties. You can configure run-time properties for the Hadoop environment in the Data Integration Service, the Hadoop connection, and in the mapping. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Hadoop connection or in the mapping. The Data Integration Service processes property overrides based on the following priorities: 1. Mapping custom properties set using infacmd ms runMapping with the `-cp` option 2. Mapping run-time properties for the Hadoop environment 3. Hadoop connection advanced properties for run-time engines 4. Hadoop connection advanced general properties, environment variables, and classpaths 5. Data Integration Service custom properties **Note:** Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.

Spark Configuration

The following table describes the connection properties that you configure for the Spark engine:

Property	Description
Spark Staging Directory	The HDFS file path of the directory that the Spark engine uses to store temporary files for running jobs. The YARN user, Data Integration Service user, and mapping impersonation user must have write permission on this directory. If you do not specify a file path, by default, the temporary files are written to the Hadoop staging directory `/tmp/SPARK_<user name>`. When you run Sqoop jobs on the Spark engine, the Data Integration Service creates a Sqoop staging directory within the Spark staging directory to store temporary files: `<Spark staging directory>/sqoop_staging`
Spark Event Log Directory	Optional. The HDFS file path of the directory that the Spark engine uses to log events.
YARN Queue Name	The YARN scheduler queue name used by the Spark engine that specifies available resources on a cluster. The name is case sensitive.
Advanced Properties	List of advanced properties that are unique to the Spark engine. The advanced properties include a list of default properties. You can configure run-time properties for the Hadoop environment in the Data Integration Service, the Hadoop connection, and in the mapping. You can override a property configured at a high level by setting the value at a lower level. For example, if you configure a property in the Data Integration Service custom properties, you can override it in the Hadoop connection or in the mapping. The Data Integration Service processes property overrides based on the following priorities: 1. Mapping custom properties set using infacmd ms runMapping with the `-cp` option 2. Mapping run-time properties for the Hadoop environment 3. Hadoop connection advanced properties for run-time engines 4. Hadoop connection advanced general properties, environment variables, and classpaths 5. Data Integration Service custom properties **Note:** Informatica does not recommend changing these property values before you consult with third-party documentation, Informatica documentation, or Informatica Global Customer Support. If you change a value without knowledge of the property, you might experience performance degradation or other unexpected results.

Troubleshooting the Big Data Management Integration with Qubole

This section contains a list of issues which you should be aware of.

The issues in this section are particular to the Big Data Management 10.2.2 HF1 SP1 integration with Qubole. To read about other known limitations that also affect some Big Data Management 10.2.2 HF1 SP1 implementations and use cases, read the Big Data Management 10.2.2 HF1 SP1 Release Notes.

Big Data Management Known Limitations

The following table describes known limitations:

Issue	Description

BDM-31950	A mapping with a Hive target with truncation and partitioning enabled fails with an error like:
	`FAILED: SemanticException [Error 10096]: Dynamic partition strict mode requires at least one static partition column.`
	Workaround: Use the Administrator tool to perform the following steps:
	1. Add the following property-value pairs to the hive_site_xml configuration set in the cluster configuration:
	- `hive.support.concurrency=TRUE`
	- `hive.enforce.bucketing=TRUE`
	- `hive.exec.dynamic.partition.mode=nonstrict`
	- `hive.txn.manager=org.apache.hadoop.hive.ql.lockmgr.DbTxnManager`
	- `hive.compactor.initiator.on=TRUE`
	- `hive.compactor.worker.threads=1`
	2. Edit both the Hive and Hadoop connections to configure the Environment SQL property with the following value:
	`set hive.exec.dynamic.partition.mode=nonstrict; set mapred.java.child.opts=-Xmx1g; set io.sort.mb=10;`
	Note: In the Hive connection, the Environment SQL property is under the Common Properties heading. In the Hadoop connection, the Environment SQL property is under the Hive Pushdown Configuration heading.

BDM-31947	A mapping that runs on the Spark engine, with an external Hive target configured to truncate target and to retain the target schema strategy, runs successfully, but the target table is not truncated. Workaround: On the **Data Object** tab of the target properties, enable the option "At run time, get data object columns from the data source."
BDM-31948	A mapping that runs on the Spark engine, with an external Hive target configured to truncate target, to retain the target schema strategy, and and target partition enabled, fails with an error like: `Error: Hive message [Error while compiling statement: FAILED: SemanticException 1:23 Need to specify partition columns because the destination table is partitioned. Error encountered near token 'truncate_<table name>'], Hive SQL state [42000]./` Workaround: On the **Data Object** tab of the target properties, enable the option "At run time, get data object columns from the data source."

Third Party Known Limitations

The following table describes third party known limitations:

Issue	Description
BDM-31946	When a Hive target is defined as a JSON SerDe target on a mapping that runs on Spark, the mapping fails with a "class not found" error. This happens because Hive fails to add the Hive catalog .jar file that enables Hive CLI. If you add the file manually, the mapping still fails due to the Spark issue listed below: Hive ticket reference number: HIVE-17624 Spark ticket reference number: SPARK-29002
BDM-31944	When a Hive target table has "APPLYNEWSCHEMA" enabled for the target schema strategy and the mapping is configured to run on Spark, the mapping fails with an error like: *SQLException: [org.apache.hive.service.cli.HiveSQLException: java.io.IOException: org.apache.hadoop.hive.ql.metadata.HiveException: java.lang.ClassCastException: org.apache.hadoop.hive.serde2.io.HiveDecimalWritable cannot be cast to org.apache.hadoop.io.IntWritable* Hive ticket reference number: HIVE-19317

Author

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